Hui Jiang

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RESEARCH EXPERTISE

Artificial Intelligence; Machine Learning; Deep Learning;
Speech Recognition; Natural Language Processing; Reasoning; Computer Vision

EDUCATION

The University of Tokyo, Tokyo, Japan – Ph.D., 1998.
University of Science and Technology of China, Hefei, China
- M. Sc. (Electrical Engineering), 1994.
- B. Eng (Electrical Engineering), 1992.

ACADEMIC APPOINTMENTS

Director 2015-Present
The Laboratory for Neural Computing and Machine Learning (NCML) Department of Electrical Engineering and Computer Science, Lassonde School of Engineering, York University.

Vector Faculty Affiliate 2018-Present
Vector Institute for Artificial Intelligence, 661 University Ave Suite 710, Toronto

iFLYTEK Endowed Professor 2015-2020
Lassonde School of Engineering, York University

Full Professor 2013-Present
Associate Professor 2007-2013
Assistant Professor 2002-2007
Department of Electrical Engineering and Computer Science, Lassonde School of Engineering, York University.

Guest Professor 2007-2018
School of Information Science and Technology, University of Science and Technology of China, Hefei, Anhui, China.

Researcher 2000-2002
Dialogue Systems Research Department, Multimedia Communication Research Lab., Bell Laboratories, Lucent Technologies Inc., Murray Hill, NJ 07974, USA.

Postdoctoral Fellow 1999-2000
Department of Electrical and Computer Engineering, University of Waterloo, Waterloo, Ontario, Canada.

Research Scientist 1998-1999
Department of Information and Communication Engineering, The University of Tokyo, Tokyo, Japan.

AWARDS AND RECOGNITIONS

- **2017 Lassonde Innovation Award** (Established Researcher)
- **2017 ACL Outstanding Paper Honour**
- **2016 IEEE Signal Processing Society (SPS) Best Paper Award**
- Winning the **top two positions** (No.1 and No.2) in TAC KBP Trilingual Entity Discovery and Linking (EDL) Track, organized by NIST in September 2016.
- Winning **first** position in the Winograd Schema Challenge on Common-sense Reasoning, held in IJCAI 2016 in New York City, July 2017, covered by **MIT Technology Review** and **Science Alert**.
- York University Research Leader, 2015-2016.
- Merit Awards, York University, 2003, 2005, 2006, 2008, 2009, 2010 and 2011.
- Monbusho fellowship, Japanese Government, 1995-1998.

SCHOLARLY AND PROFESSIONAL ACADEMIC ACTIVITIES

- Session Chair, Interspeech, San Francisco, September 2016
- Associate Editor, IEEE Transactions on Audio, Speech, and Language Processing, 2009-2013.
- Associate Editor (TPC member) of the Signal and Multimedia Processing Symposium at IEEE CCECE 2012, Montreal, Quebec, Canada, May 2012.
- Technical Program Committee, 7th International Symposium on Chinese Spoken Language Processing, Tainan, Taiwan, December 2010.
- Technical Program Committee, Asia Pacific Signal and Information Processing Association (APSIPA), 2009-2010.

- Invited external grant reviewer, American Institute of Biological Sciences (AIBS), 2009.
- Technical Program Chair, 6th International Symposium on Chinese Spoken Language Processing, Kunming, China, December 2008.
- Plenary Sessions Chair, Organizing Committee, 2004 IEEE International Conference on Acoustic, Speech and Signal Processing (ICASSP-2004), Montreal, Canada, May 2004.
- Technical Committee Member, 2004 International Symposium on Chinese Spoken Language Processing (ISCSLP'2004), Hong Kong, China, December 2004.
- Technical Committee Member, 2006 International Symposium on Chinese Spoken Language Processing (ISCSLP'2006), Singapore, December 2006.
- Technical Committee Member, 2006 IEEE Symposium on Signal Processing and Information Technology, held in Vancouver, Canada on August 27-30, 2006.
- Associated Editor of The Journal of Advances in Multimedia, Hindawin Publishing Corporation since 2006.
- Panel Member, 2006-2007 Ontario Graduate Scholarship Program, Ministry of Training, Ontario.
- Senior member of IEEE.
- Professional Engineer of PEO (Professional Engineers Ontario) since 2003.
- Reviewers of IEEE Transactions, IEEE signal processing letters, Speech Communication, IEE Journals, ICASSP, Interspeech, ACL, EMNLP, NAACL, NIPS.
- Invited Reviewer for NSERC (Discovery grants & Strategic grants).

EXTERNAL RESEARCH FUNDING

- 2018-2021: NSERC, $120,000, *Exploring New Neural Computing Models for Natural Language Understanding,* Discovery Accelerator Supplements (DAS), PI.
- 2018-2024: NSERC, $64,000/yr, *Exploring New Neural Computing Models for Natural Language Understanding,* Discovery, PI.
- 2015-2020: **iFLYTEK**, **$1.5 M**, *Neural Computing for Perception and Cognition*, industry donation, PI.
- 2014: NSERC, $25,000, *Investigate Deep Learning Methods for Large-Scale Financial Event Modelling*, Engage, PI,
- 2013-2017: NSERC, $25,000/yr, 5-yr grant, *Large-Scale Discriminative Modelling for Data-Intensive Speech and Language Processing*, Discovery, PI.

- 2008-2012: NSERC, $28,000/yr, 5-yr grant, *Margin-based Discriminative Learning of Hidden Markov Models for Speech Recognition*, The Discovery Funds, PI.
- 2007-2011: **NIH (US)**, US$50,490/yr, 5-yr grant, *Audio Processing for Cochlear Implants,* Research Grant, PI.
- 2006-2007: Cybernetics InfoTech Inc. (US), *Fixed-Target Voice Conversion*, $28,000, Industry Research Grant, PI.
- 2003-2004: PRECARN/IRIS, *A Spoken Dialogue System for Talking with Robots*,$32,000, The Emerging Opportunity Funds, PI.
- 2003--2007: NSERC, $24,000/yr, 5-yr grant, *Voice Interface for Wireless Personal Hand-held Devices*, The Discovery Funds, PI.

INVITED PRESENTATIONS

- April 2019, *Understanding Deep Learning in Theory*, **Vector Institute**, Toronto
- October 2017, *A New General Deep Learning Approach for Natural Language Processing*, **University of Montreal**, Canada. (Invited and hosted by Prof. Yoshua Bengio)
- December 2016, *A New General Deep Learning Approach for Natural Language Processing* , **Harbin Institute of Technology**, China. (Invited and hosted by Prof. Ting Liu)
- November 2016, *Neural Models for Entity Discovery and Linking*, **Text Analysis Conference(TAC) 2016 Workshop**, Gaithersburg, Maryland USA.
- November 2015, *A New Framework to Learn Neural Networks: Hybrid Orthogonal Projection and Estimation*, Department of Electrical Engineering, **Tsinghua University**, Beijing, China. (Invited and hosted by Prof. Ji Wu)
- August 2015, *A New Framework to Learn Neural Networks: Hybrid Orthogonal Projection and Estimation*, **Microsoft Research Asia**, Beijing, China. (Invited and hosted by Dr. Qiang Huo)
- November 2013, *Why Deep Neural Networks Work For Speech,* Department of Electrical and Computer Engineering, **Ryerson University**, Toronto. (Invited and hosted by Prof. Xiao-Ping Zhang)
- October 2013, *Why DNN Works for Speech and How to Make it More Efficient,* **IBM Waston Research Center**, Yorktown Heights, New York. (Invited and hosted by Dr. Xiaodong Cui)
- February 2013, *Why DNN Works for Acoustic Modelling in Speech Recognition*, Department of Computer Science, **University of Toronto**. (Invited and hosted by Prof. Gerald Penn)

- October 2010, *Discriminative Training of Generative Models with its Application on Speech and Language Processing*, **University of Science and Technology of China**, Hefei, China. (Invited and hosted by Prof. Li-Rong Dai)
- May 2009, *Convex Optimization for Discriminative Training of HMMs in Speech Recognition*, **National Cheng-Kung University**, Taiwan. (Invited and hosted by Prof. Jen-Tzung Chien)
- November 2008, *A New Framework for Discriminative Learning of Generative Models*, **University of Science and Technology of China**, Hefei, China. (Invited and hosted by Prof. Ren-Hua Wang)
- October 2007, *Large Margin Estimation of Hidden Markov Models for Speech Recognition*, an invited plenary talk in the **9th National Conference on Man-Machine Communication**, Huangshan, China.
- August 2006, *Solving Large Margin Hidden Markov Models via Semi-definite Programming*, **Microsoft Research Asia**, Beijing, China. (Invited and hosted by Dr. Frank Soong)
- June 2006, *Acoustic Modeling For Speech Recognition: From Maximum Likelihood (ML) toDiscriminative Training (DT) and Beyond*, an invited tutorial talk in the **9th Western Pacific Acoustics Conference**, Seoul, Korea.
- February 2006, *Large Margin HMMs for Speech Recognition*, **Microsoft Research**, Redmond, Washington. (Invited and hosted by Dr. Li Deng)
- January 2006, *Model-Based Speech Processing: Recognition, Enhancement and Separation*, Zarlink Semiconductor Inc., Ottawa, Ontario. (Invited and hosted by Dr. Gary Jin)
- February 2005, *Large-Margin HMM Estimation for Speech Recognition*, **IBM Waston Research Center**, Yorktown Heights, New York. (Invited and hosted by Dr. Olivier Siohan)
- April 2004, *An In-Search Data Selection Algorithm and Its Applications*, **Microsoft Research Asia**, Beijing, China. (Invited and hosted by Dr. Ming Zhou)
- November 2003, *Confidence Measures: How Much We Can Trust Our Speech Recognizers,* **University of Toronto**. (Invited and hosted by Prof. Parham Aarabi)
- April 2000, *Voice User Interface for In-Car Applications*, **Motorola Inc.**, Austin, Taxes. (Invited and hosted by Dr. Jeremy Lao)

PUBLICATIONS

(my supervised/co-supervised students and hosted postdocs in bold)

BOOK AND CHAPTERS

1. H. Jiang, *Machine Learning Fundamentals: A Concise Introduction*, Cambridge University Press, 2021.

2. **P. Hu**, H. Jiang and A. Emili, ``Incorporating correlations among gene ontology terms into predicting protein functions,'' *Invited chapter for Ontology learning and knowledge discovery using the web: challenges and recent advances (edited by W Wong, W Liu, M Bennamoun)*, IGI Global Press, 2011.

3. **D. Wu**, **B. Li** and H. Jiang, ``Normalization and Transformation Techniques for Robust Speaker Recognition," *Speech Recognition, Technologies and Applications*, edited by F. Mihelic and J. Zibert, November 2008.

4. **P. Hu**, C. Chung, H. Jiang and A. Emili, ``Chapter 13: Bioinformatics applications: predicting protein subcellular localization by applying machine learning classifiers to proteomic expression profiles", *Introduction to Bioinformatics: A concept based approach, (edited by V Mathura and P Kangueane)*, pp.163-174, Springer-Verlag Press, October 2008.

5. H. Jiang and **X. Li**, ``A General Approximation-Optimization Approach to Large Margin Estimation of HMMs," *Speech Recognition and Synthesis (edited by V. Kodic)*, 2007.

REFERRED JOURNAL PAPERS

6. **S. Zhang**, C. Liu, H. Jiang, S. Wei, L. Dai, Y. Hu, "Non-Recurrent Neural Structure for Long-Term Dependency," *IEEE/ACM Trans. on Audio, Speech and Language Processing*, pp. 871-884, Vol. 25, No. 4, April 2017.

7. **S. Zhang**, H. Jiang, L. Dai, ``Hybrid Orthogonal Projection and Estimation (HOPE): A New Framework to Learn Neural Networks," *Journal of Machine Learning Research (JMLR)*, 17(37):1−33, 2016.

8. **S. Xue**, H. Jiang, L. Dai, Q. Liu, ``Speaker Adaptation of Hybrid NN/HMM Model for Speech Recognition Based on Singular Value Decomposition," *Journal of Signal Processing Systems*, pp.175-185, Vol 82, Issue 2, February 2016.

9. **P. Zhou**, H. Jiang, L. Dai, Y. Hu and Q. Liu, ``State-Clustering based Multiple Deep Neural Networks Modelling Approach for Speech Recognition," *IEEE/ACM Trans. on Audio, Speech and Language Processing*, pp.631-642, Vol. 23, No. 4, April 2015.

10. **S. Xue**, **O. Abdel-Hamid**, H. Jiang, L. Dai and Q. Liu, ``Fast Adaptation of Deep Neural Network based on Discriminant Codes for Speech Recognition," *IEEE/ACM Trans. on Audio, Speech and Language Processing*, pp.1713-1725, Vol. 22, No. 12, December 2014. (**Google Scholar citation is 131 as of July 2020**)

11. H. Jiang, **Z. Pan** and **P. Hu**, ``Discriminative Learning of Generative Models: Large Margin Multinomial Mixture Models for Text Categorization," *Pattern Analysis and Applications*, *pp.535-551, Volume 18, Issue 3,* August 2015.

12. **O. Abdel-Hamid**, A.-R. Mohamed, H. Jiang, L. Deng, G. Penn and D. Yu, ``Convolutional Neural Networks for Speech Recognition," *IEEE/ACM Trans. on Audio, Speech and Language Processing*, pp.1533-1545, Vol. 22, No. 10, October 2014. (**Google Scholar citation is 1333 as of July 2020**) **(2016 IEEE Signal Processing Society Best Paper Award)**

13. E. P. Wilkinson, **O. Abdel-Hamid**, J. J. Galvin, H. Jiang and Q.-J. Fu, ``Voice Conversion in Cochlear Implantation", *The Laryngoscope*, Vol. 123, Issue S13, pp.S29-S43, January 2013.

14. **P. Hu**, S. Bull and H. Jiang, ``Gene network modular-based classification of microarray samples", *BMC Bioinformatics, 13(Suppl 10):S17*, 2012.

15. **C. Liu**, Y. Hu, L.-R. Dai and H. Jiang, ``Trust Region-Based Optimization for Maximum Mutual Information Estimation of HMMs in Speech Recognition", *IEEE Trans. on Audio, Speech and Language Processing*, pp.2474 - 2485, Vol. 19, No. 8, November 2011.

16. **J. Du**, Y. Hu and H. Jiang, ``Boosted Mixture Learning of Gaussian Mixture Hidden Markov Models Based on Maximum Likelihood for Speech Recognition", *IEEE Trans. on Audio, Speech and Language Processing*, pp.2091-2100, Vol. 19, No. 7, September 2011.

17. **D. Wu**, **Y. Yin** and H. Jiang, ``Large Margin Estimation of Hidden Markov Models with Second Order Cone Programming for Speech Recognition", *IEEE Trans. on Audio, Speech and Language Processing*, pp.1652-1664, Vol. 19, No. 6, August 2011.

18. H. Jiang, ``Discriminative Training for Automatic Speech Recognition: A Survey," *Computer and Speech, Language,* pp.589-608, Vol. 24, Issue 4, October 2010. (**Google Scholar citation is 102 as of July 2020**)

19. H. Jiang and **X. Li**, ``Parameter Estimation of Statistical Models Using Convex Optimization: An Advanced Method of Discriminative Training for Speech and Language Processing", *IEEE Signal Processing Magazine*, pp. 115-127, Vol. 27, No. 3, May 2010.

20. **P. Hu**, H. Jiang and A. Emili, ``Predicting protein functions by relaxation labelling protein interaction network," *BMC Bioinformatics 2010, 11(Suppl 1):S64*, January 2010.

21. P. Liu, **C. Liu**, H. Jiang, F. Soong and R.-H. Wang,``A Constrained Line Search Optimization for Discriminative Training of HMMs," *IEEE Trans. on Audio, Speech and Language Processing*, pp.900-909, Vol. 16, No. 5, July 2008.

22. **X. Li** and H. Jiang, ``Solving Large Margin HMM Estimation via Semidefinite Programming," *IEEE Trans. on Audio, Speech and Language Processing*, pp.2383-2392, Vol. 15, No. 8, November 2007.

23. M. Afify, **X. Li** and H. Jiang, ``Statistical Analysis of Minimum Classification Error Learning for Gaussian and Hidden Markov Model Classifiers," *IEEE Trans. on Audio, Speech and Language Processing*, pp.2405-2417, Vol. 15, No.8, November 2007.

24. **P. Hu**, H. Jiang and A. Emili, ``Comparison of computational strategies for multi-label prediction of protein subcellular localizations," *International Journal of Information Technology and Intelligence Computing*, Vol. 2, No.2, 2007.

25. **G. Shi**, P. Aarabi, and H. Jiang, ``Phase-based Dual-MicrophoneSpeech Enhancement Using a Prior Speech Model," *IEEE Trans. on Audio, Speech and Language Processing*, pp. 109-118, Vol. 15, No. 1, January 2007.

26. Y. Tian, J.-L. Zhou, H. Lin, and H. Jiang, ``Tree-based Covariance Modelling of Continuous Density Hidden Markov Models," *IEEE Trans. on Audio, Speech and Language Processing*, pp.2134-2146, Vol. 14, No. 6, November 2006.

27. H. Jiang, **X. Li**, and **C. Liu**, ``Large Margin Hidden Markov Models for Speech Recognition," *IEEE Trans. on Audio, Speech and Language Processing*, pp. 1584-1595, Vol. 14, No. 5, September 2006. (**Google Scholar citation is 158 as of July 2020**)

28. H. Jiang, F. Soong, C.-H. Lee, ``A dynamic in-search data selection method with its applications to acoustic modelling and utterance verification," *IEEE Trans. on Speech and Audio Processing*, pp.945-955, Vol. 13, No.5, September 2005.

29. H. Jiang, ``Confidence Measures for Speech Recognition: A Survey", *Speech Communication*, Vol. 45, No. 4, pp. 455-470, April 2005. (**Google Scholar citation is 467 as of July 2020**)

30. M. Afify, F. Liu, H. Jiang, and O. Siohan, ``A new verification-based fast match for large vocabulary speech recognition," *IEEE Trans. on Speech and Audio Processing*, pp.546-553, Vol. 13, No.4, July 2005.

31. H. Jiang, C.-H. Lee, ``A new approach to utterance verification based on neighbourhood information in model space," *IEEE Trans. on Speech and Audio Processing*, pp.425-434, Vol. 11, No.5, September 2003.

32. H. Jiang and L. Deng, ``A robust compensation strategy against extraneous acoustic variations for spontaneous speech recognition," *IEEE Trans. on Speech and Audio Processing*, pp.9-17, Vol. 10, No.1, January, 2002.

33. H. Jiang and L. Deng, ``A Bayesian approach to the verification problem: — applications to speaker verification," IEEE Trans. on Speech and Audio Processing, pp.874-884, Vol. 9, No.8, November 2001.

34. H. Jiang, K. Hirose, and Q. Huo, ``A minimax search algorithm for robust continuous speech recognition," *IEEE Trans. on Speech and Audio Processing*, pp. 688-694, Vol. 8, No. 6, November 2000.

35. H. Jiang, K. Hirose, and Q. Huo, ``Improving Viterbi Bayesian predictive classification via sequential Bayesian learning in robust speech recognition," *Speech Communication*, pp. 313-326, Vol 28, Issue 4, August 1999. (with the invitation of Editorial Board)

36. H. Jiang, K. Hirose, and Q. Huo, ``Robust speech recognition based on Bayesian prediction approach", *IEEE Trans. on Speech and Audio Processing*, pp. 426-440, Vol. 7, No.4, July 1999. (**Google Scholar citation is 76 as of Jan 2020**)

REFERRED CONFERENCE PAPERS

37. **C. Wang**, J. Gaspers, Q. Do and H. Jiang, "Exploring Cross-Lingual Transfer Learning with Unsupervised Machine Translation." *Findings of the Association for Computational Linguistics (ACL)*, August 2021.

38. **C. Wang** and H. Jiang, "Explicit Utilization of General Knowledge in Machine Reading Comprehension", *Proc. of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, July 2019.

39. **C. Wang** and H. Jiang, "The Lower The Simpler: Simplifying Hierarchical Recurrent Models", *Proc. of North American Chapter of the Association for Computational Linguistics (NAACL)*, June 2019.

40. K. Jiang, **D. Wu** and H. Jiang, "FreebaseQA: A New Factoid QA Data Set Matching Trivia-Style Question-Answer Pairs with Freebase," *Proc. of North American Chapter of the Association for Computational Linguistics (NAACL)*, June 2019.

41. **K. Joseph** and H. Jiang, "Content based News Recommendation via Shortest Entity Distance over Knowledge Graphs," *Proc. of WWW2019 Workshop On Knowledge Graph Technology And Applications*, May 2019.

42. **S. Watcharawittayakul**, **M. Xu** and Hui Jiang, "Dual Fixed-Size Ordinally Forgetting Encoding (FOFE) for Competitive Neural Language Models," *Proc. of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2018)*, October, 2018. ***(oral acceptance rate of 9.5%)***

43. **H. Pan** and H. Jiang, ``Learning Convolutional Neural Networks using Hybrid Orthogonal Projection and Estimation,'' *Proc. of the 9th Asian Conference on Machine Learning (ACML)*, November 2017. ***(acceptance rate of 23.8%)***

44. **J. Sanu, M. Xu,** H. Jiang and **Q. Liu**, ``Word Embeddings based on Fixed-Size Ordinally Forgetting Encoding,'' *Proc. of the Conference on Empirical Methods in Natural Language Processing (EMNLP 2017), September, 2017.* ***(acceptance rate of 18.4%)***

45. **Q. Liu**, H. Jiang, **A. Evdokimov**, Z. Ling, X. Zhu, S. Wei and Y. Hu, ``Cause-Effect Knowledge Acquisition and Neural Association Model for Solving A Set of Winograd Schema Problems,'' *Proc. of International Joint Conference on Artificial Intelligence (IJCAI 2017)*, August 2017. **(acceptance rate is 26%)**

46. **M. Xu**, H. Jiang and **S. Watcharawittayakul**, ``A Local Detection Approach for Named Entity Recognition and Mention Detection,'' *Proc. of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, July 2017. ***(one of Outstanding Papers selected by ACL2017)*** **(Google Scholar citation is 60 as of July 2020)**

47. Q. Chen, X. Zhu, Z. Ling, S. Wei, H. Jiang and D. Inkpen, ``Enhanced LSTM for Natural Language Inference,'' *Proc. of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, July 2017. **(Google Scholar citation is 465 as of July 2020)**

48. **H. Pan** and H. Jiang, ``A Fast Method for Saliency Detection by Back-propagating a Convolutional Neural Network and Clamping Its Partial Outputs,'' *Proc. of International Joint Conference on Neural Networks (IJCNN)*, May 2017.

49. **S. Zhang**, H. Jiang, S. Xiong, S. Wei, L. Dai, ``Compact Feedforward Sequential Memory Networks for Large Vocabulary Continuous Speech Recognition," *Proc. of Interspeech*, September 2016.

50. **Q. Chen**, X. Zhu, Z. Ling, S. Wei and H. Jiang,``Distraction-based Neural Networks for Modelling Documents," *Proc. of International Joint Conference on Artificial Intelligence (IJCAI)*, July 2016. **(acceptance rate is 25%)** **(Google Scholar citation is 106 as of July 2020)**

51. **Y. Peng** and H. Jiang, ``Leverage Financial News to Predict Stock Price Movements UsingWord Embeddings and Deep Neural Networks," *Proc. of North American Chapter of the Association for Computational Linguistics (NAACL)*, June 2016. (**acceptance rate is 29%**) (**Google Scholar citation is 47 as of July 2020**)

52. **Q. Liu**, W. Guo, Z. Ling, H. Jiang and Y. Hu, ``Intra-Topic Variability Normalization based on Linear Projection for Topic Classification," *Proc. of North American Chapter of the Association for Computational Linguistics (NAACL)*, June 2016. (**acceptance rate is 29%**)

53. **S. Zhang**, H. Jiang, S. Wei, L. Dai, ``Rectified Linear Neural Networks with Tied-Scalar Regularization for LVCSR," *Proc. of Interspeech*, September 2015.

54. **S. Zhang**, H. Jiang, M. Xu, J. Hou, L. Dai, ``The Fixed-Size Ordinally-Forgetting Encoding Method for Neural Network Language Models," *Proc. of the 53th Annual Meeting of the Association for Computational Linguistics (ACL 2015)*, July, 2015. (**acceptance rate is 22.3%**) (**Google Scholar citation is 61 as of July 2020**)

55. Z. Chen, X. Zhu, W. Lin, **Q. Chen,** S. Wei, X. Chen, H. Jiang, ``Revisiting Word Embedding for Contrasting Meaning." *Proc. of The 53th Annual Meeting of the Association for Computational Linguistics (ACL 2015)*, July, 2015. (**acceptance rate is 25%**)

56. **Q. Liu**, H. Jiang, S. Wei, Z.-H. Ling, Y. Hu,``Learning Semantic Word Embeddings based on Ordinal Knowledge Constraints," *Proc. of The 53th Annual Meeting of the Association for Computational Linguistics (ACL 2015)*, July, 2015. (**acceptance rate is 25%**) (**Google Scholar citation is 120 as of July 2020**)

57. **H. Pan**, H. Jiang, ``Annealed Gradient Descent for Deep Learning," *Proc. of 31th Conference on Uncertainty in Artificial Intelligence (UAI 2015)*, July 2015. (**acceptance rate is 34%**)

58. **S. Zhang**, H. Jiang, L. Dai, ``The New HOPE Way to Learn Neural Networks", *Proc. of Deep Learning Workshop at ICML 2015*, July 2015.

59. **S. Xue**, H. Jiang, L. Dai, Q. Liu, ``Unsupervised Speaker Adaptation of Deep Neural Network based on the combination of speaker codes and singular value decomposition for Speech Recognition," *Proc. of of IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'15)*, April 2015.

60. **S. Xue**, H. Jiang, L. Dai, ``Speaker Adaptation of Hybrid NN/HMM Model for Speech Recognition Based on Singular Value Decomposition" *Proc. of International Symposium on Chinese Spoken Language Processing (ISCSLP'2014)*, September 2014.

61. C. Kong, **S. Xue**, J. Gao, W. Guo, L. Dai and H. Jiang, ``Speaker Adaptive Bottleneck Features Extraction for LVCSR Based on Discriminative Learning of Speaker Codes," *Proc. of International Symposium on Chinese Spoken Language Processing (ISCSLP'2014)*, September 2014.

62. **S. Zhang**, **Y. Bao**, **P. Zhou**, H. Jiang and L. Dai, ``Improving deep neural networks for LVCSR using dropout and shrinking structure," *Proc. of IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'14)*}, May 2014.

63. **S. Xue**, **O. Abdel-Hamid**, H. Jiang and L. Dai, ``Direct adaptation of hybrid DNN/ HMM model for fast speaker adaptation in LVCSR based on speaker code," *Proc. of IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'14)*, May 2014. (**Google Scholar citation is 66 as of July 2020**)

64. **P. Zhou**, L. Dai and H. Jiang, ``Sequence training of multiple deep neural networks for better performance and faster training speed," *Proc. of IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'14)*, May 2014.

65. **O. Abdel-Hamid**, L. Deng, D. Yu and H. Jiang, ``Deep Segmental Neural Network for Automatic Speech Recognition," *Proc. of Interspeech*, Lyon, France, 2013.

66. **O. Abdel-Hamid** and H. Jiang, ``Rapid and Effective Speaker Adaptation of Convolutional Neural Network Based Models for Speech Recognition," Proc. of Interspeech, Lyon, France, 2013. (**Google Scholar citation is 56 as of July 2020**)

67. **Y. Bao**, H. Jiang, L. Dai, **C. Liu**,``Incoherent Training of Deep Neural Networks to De-correlate Bottleneck Features for Speech Recognition," *Proc. of 2013 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP'13)*, Vancouver, Canada.

68. **O. Abdel-Hamid** and H. Jiang, ``Fast Speaker Adaptation Of Hybrid NN/HMM Model for Speech Recognition based on Discriminative Learning of Speaker Code," Proc. of IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP), Vancouver, Canada, 2013. (**Google Scholar citation is 209 as of July 2020**)

69. **P. Zhou**, **C. Liu**, Q. Liu, L. Dai and H. Jiang, ``A cluster-based multiple deep neural networks method for large vocabulary continuous speech recognition," *Proc. of 2013 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, Vancouver, Canada, 2013.

70. J. Pan, **C. Liu**, Z. Wang, Y. Hu and H. Jiang, ``Investigation of Deep Neural Networks for Large Vocabulary Continuous Speech Recognition", *Proc. of International Symposium on Chinese Spoken Language Processing (ISCSLP)*, Hong Kong, December 2012. (**Google Scholar citation is 97 as of Jan 2020**)

71. **Y. Bao**, H. Jiang, **C. Liu**, Y. Hu and L. Dai,``Investigation on dimensionality reduction of concatenated features with deep neural network for LVCSR systems," *Proc. of IEEE International Conference on Signal Processing (ICSP)*, 2012.

72. **O. Abdel-Hamid**, A. Mohamed, H. Jiang, G. Penn, ``Applying Convolutional Neural Networks Concepts to Hybrid NN-HMM Model for Speech Recognition," *Proc. of IEEE International Conference on Acoustic, Speech, Signal Processing (ICASSP)*, Kyoto, Japan, March 2012. (**Google Scholar citation is 831 as of July 2020**)

73. **P. S. Jafari**, H.-Y. Kang, X. Wang, Q.-J. Fu and H. Jiang, ``Phase-sensitive Speech Enhancement for Cochlear Implant Processing," *Proceedings of IEEE International Conference on Acoustic, Speech, and Signal Processing (ICASSP)*, Prague, Czech, May 2011.

74. **P. Hu** and H. Jiang, ``Classification of high-throughput data using correlation-shared gene clusters," *Proceedings of International Conference on Bioinformatics and Computational Biology (BIOCOMP)*, Las Vegas, USA, July 2011.

75. **P. Hu**, S. Bull and H. Jiang, ``Gene network models-based linear discriminant analysis of microarray expression data", *Proceedings of 7th International Symposium on Bioinformatics Research and Applications (ISBRA'2011) , LNBI 6647, pp. 286-296, Springer-Verlag Berlin Heidelberg, edited by J. Chen, J. Wang and A. Zelikovsky*, May 2011.

76. **J. Du**, Y. Hu and H. Jiang, ``Boosted Mixture Learning of Gaussian Mixture HMMs for Speech Recognition", *Proc. of Interspeech*, Makuhari, Japan, September 2010.

77. **V. Magdin** and H. Jiang, ``Large Margin Estimation of N-gram Language Models for Speech Recognition via Linear Programming", *Proceedings of IEEE International Conference on Acoustic, Speech, and Signal Processing (ICASSP)*, Dallas, Taxes, March 2010.

78. **C. Liu**, Y. Hu, H. Jiang and L.-R. Dai, ``A Bounded Trust Region Optimization for Discriminative Training of HMMs in Speech Recognition", *Proceedings of IEEE International Conference on Acoustic, Speech, and Signal Processing (ICASSP)*, Dallas, Taxes, March 2010.

79. **V. Magdin** and H. Jiang, ``Discriminative Training of N-gram Language Models for Speech Recognition via Linear Programming", *Proc. of IEEE Workshop on Automatic Speech Recognition and Understanding (ASRU)*, Merano, Italy, December 2009.

80. **P. Hu**, H. Jiang and A. Emili, ``A topology-sharing based method for protein function prediction via analysis of protein functional association networks", *Proceedings of IEEE International Conference on Bioinformatics and Biomedicine Workshops*, pp.243-248, Washington DC, USA, December 2009.

81. **D. Wu**, **B. Li** and H. Jiang, ``Maximum Mutual Information Estimation via Second Order Cone Programming for Large Vocabulary Continuous Speech Recognition," *Proc. of Interspeech*, Brighton, UK, September 2009.

82. **D. Wu**, **B. Li**, H. Jiang and Q. Fu, ``Online Model Adaptation for Voice Conversion using Model-based Speech Synthesis Technique," *Proc. of Interspeech*, Brighton, UK, September 2009.

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PATENTS

1. US patent (US 20140288928 A1), *System and method for applying a convolutional neural network to speech recognition,* November 2015. (successfully assigned to a well-funded startup in Canada on Sep 2017)

2. US patent (US 8239332 B2), *Constrained line search optimization for discriminative training of HMMs*, August 2012.

GRADUATE SUPERVISION

Supervised: Master Theses (15 completed, 3 in progress); PhD Theses (3 completed, 3 in progress); Postdoc (3 completed, 0 in progress)

Co-supervised: Master Theses (2 completed, 0 in progress); PhD Theses (7 completed, 2 in progress)

POSTDOC SUPERVISION

1. Dalei Wu: supervisor (03/08 – 02/10), speech recognition, CSE, York

2. Baojie Li: supervisor (01/07 – 12/08), speech recognition, CSE, York

3. Chaojun Liu: supervisor (03/04 – 12/04), speech recognition, CS, York

PH.D. SUPERVISION

1. Zihuan Zhang: supervisor (09/2019-Present), machine learning, EECS, York

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3. Chao Wang: supervisor (09/16 – Present), machine learning, EECS, York
4. Feng Wei: supervisor (05/16 – 04/19), machine learning, EECS, York
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7. Shiliang Zhang: co-supervisor (01/13 – 06/17), speech recognition, USTC & York
8. Shaofei Xue: co-supervisor (01/12 – 06/15), speech recognition, USTC & York
9. Ossama Abdel-Hamid: supervisor (01/09 – 12/14), speech processing, CSE, York
10. Pan Zhou, co-supervisor (09/11 – 06/15), speech recognition, USTC & York
11. Yebo Bao: co-supervisor (09/10 – 06/14), speech recognition, USTC & York
12. Pingzhao Hu: supervisor (09/05 – 04/12), machine learning & bioinformatics, CSE, York
13. Cong Liu: co-supervisor (09/06 – 06/10), speech recognition, USTC & York
14. Vladimir Magdin: supervisor (01/10 – 12/11), machine learning, CSE, York
15. Xinwei Li: supervisor (09/05 – 04/10), speech recognition, CS, York
16. Pengfei Liu: supervisor (09/04 – 05/06), speech recognition, CS, York
17. Guangji Shi: co-supervisor (09/03 – 12/05), speech recognition, U of T & York

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1. Kasra Ahmadi: supervisor (09/20 — Present), Machine Learning, EECS, York
2. Behnam Asadi: supervisor (09/19 — Present), Machine Learning, EECS, York
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5. Sedtawut Watcharawittayakul: supervisor (01/17 – 12/19), Machine Learning for NLP, EECS, York
6. Dekun Jack Wu: supervisor (09/16 – 04/19), machine learning, EECS, York
7. Hong Zhang: supervisor (09/12 – 04/18), machine learning, EECS, York
8. Mingbin Xu: supervisor (01/15 – 09/17), machine learning, EECS, York
9. Yuping Lin: supervisor (01/15 – 08/17), machine learning, EECS, York

10. Rohallah Soltani: supervisor (01/14 – 04/16), machine learning, EECS, York

11. Yangtuo Peng: supervisor (09/13 – 02/16), machine learning, EECS, York

12. Poorya Seyed Jafari: supervisor (09/07 – 04/10), speech processing, CSE, York

13. Vladimir Magdin: supervisor (01/08 – 09/09), machine learning, CSE, York

14. Zhenyu Pan: supervisor (01/07 – 12/08), text categorization, CSE, York

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16. Xinwei Li: supervisor (01/04 – 08/05), speech recognition, CS, York

17. Pengfei Liu: supervisor (03/03 – 08/04), language processing, CS, York

18. Qi Wang: supervisor (07/02 – 10/04), speech recognition, CS, York

19. Koki Sasaki: co-supervisor (04/98 – 03/00), language modelling, Univ. of Tokyo

UNIVERSITY SERVICES

PROGRAM DIRECTOR/COORDINATOR

- 2017-18: Chair of Computer Science Search Committee

- 2012-13: Vice-Chair of Department of Electrical Engineering and Computer Science, Computer Engineering (CE) Program Director; Chair of Computer Engineering Committee; Chair of Electrical Engineering Search Committee (two power positions)

- 2003-04: Computer Engineering (CE) Program Coordinator, Chair of Computer Engineering Steering Committee (responsible for CE curriculum design and development, strategic planning of CE program, CE program accreditation preparation)

OTHER COMMITTEE WORK

- Departmental Executive Committees, Departmental Curriculum Committees, Departmental Search Committees, Graduate Admission Committees, Graduate Executive Committees, Graduate Computer Engineering Committee, Departmental Adjudicating Committees, File Preparation Committees, Computer Engineering Committee, FSE T&P Senate Review Committee, Departmental Internship Committee, Engineering Curriculum Committee, Engineering Program Committee (EPC)

(Updated on June. 30, 2025)